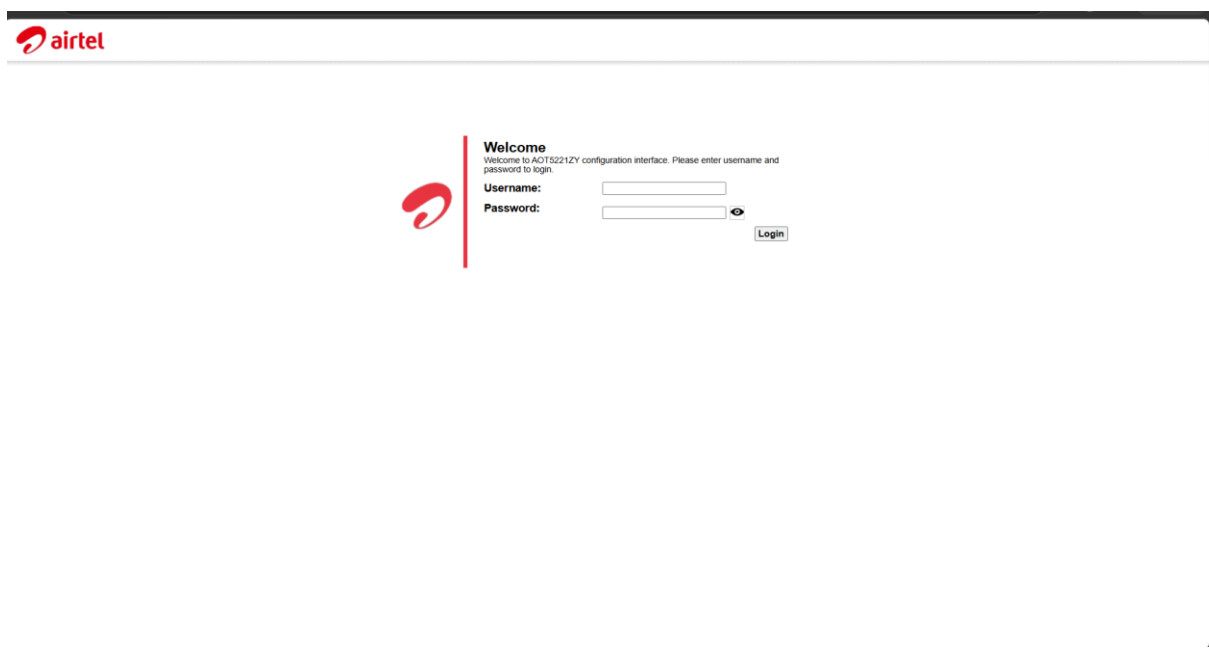


DAY-3

CONFIGURING BASIC NETWORK HARDWARE THEORY

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.



2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

```
Wireless LAN adapter Wi-Fi:
    Connection-specific DNS Suffix . : 
    IPv6 Address. . . . . : 2401:4900:8fed:2bf7:76dd:c960:5e24:9f1f
    Temporary IPv6 Address. . . . . : 2401:4900:8fed:2bf7:8dd5:2345:d835:7d88
    Link-local IPv6 Address . . . . . : fe80::6bfb:51e6:3b98:a496%8
    IPv4 Address. . . . . : 192.168.1.119
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1633:75ff:feed:45b3%8
                                192.168.1.1
```

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Remember to revert to automatic/DHCP after testing to avoid network issues.

1. Click Start → Settings.
2. Select Network & Internet → Wi-Fi.
3. Click the Wi-Fi network you are currently connected to.
4. Scroll to IP assignment and click Edit.
5. Change Automatic (DHCP) to Manual.
6. Turn IPv4 on.

7. Enter values like these, using your router's actual details:
8. Click Save.

Internet Protocol Version 4 (TCP/IPv4) Properties

General

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☐ Obtain an IP address automatically

☒ Use the following IP address:

IP address: 192 . 168 . 1 . 240

Subnet mask: 255 . 255 . 255 . 0

Default gateway: 192 . 168 . 1 . 1

☐ Obtain DNS server address automatically

☒ Use the following DNS server addresses:

Preferred DNS server: . . .

Alternate DNS server: . . .

☐ Validate settings upon exit

Advanced...

OK Cancel

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
IPv6 Address. . . . . : 2401:4900:8fed:2bf7:76dd:c960:5e24:9f1f
Temporary IPv6 Address. . . . . : 2401:4900:8fed:2bf7:1171:a874:8684:2c58
Link-local IPv6 Address . . . . . : fe80::6bfb:51e6:3b98:a496%8
IPv4 Address. . . . . : 192.168.1.240
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::1633:75ff:feed:45b3%8
                          192.168.1.1
```

4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

What happens if two devices are assigned the same IP address?

If two devices on the same network end up with identical IP addresses (an IP conflict), network communication for both becomes unreliable or fails completely. Packets intended for one device can be delivered to the other (or dropped), resulting in intermittent connectivity, “destination unreachable” errors, or total loss of internet/local network access until the conflict is resolved (usually by renewing DHCP leases or assigning a different static IP).